Isaac Ngui

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Research interests: Extended reality and hands-free interfaces that let people specify, demonstrate, and correct robot behavior, built on motion planning and robot learning from demonstration.

Education

University of Illinois Urbana-Champaign Sept. 2020 – Present
Ph.D. in Computer Science (expected May 2027) *Champaign, IL*
- Advisor: Prof. Nancy M. Amato · GPA: 3.95/4.0 · GEM University Fellow

University of Wisconsin–Milwaukee Sept. 2016 – May 2020
Bachelor of Science in Computer Science, Minor in Mathematics *Milwaukee, WI*
- *Cum Laude*

Research Experience

Graduate Research Assistant Feb. 2021 – Present
Parasol Lab, University of Illinois Urbana-Champaign · Advisor: Prof. Nancy M. Amato *Champaign, IL*
- **Extended reality for robotics** (Feb. 2023 – Present): Designing an extended reality (XR) ecosystem for interacting with robots. Built and open-sourced ERUPT (ICAT-EGVE 2026), which lets users specify and edit motion plans, reconfigure the environment, and replan interactively through MoveIt, and an XR system for collecting robot demonstrations (IEEE VR 2025 poster). Now extending the ecosystem to learning from demonstration (corrective demonstrations and behavior cloning), and investigating how XR's ability to alter the virtual world can improve robot learning and make it easier for people to express demonstrations and preferences.
- **Assistive robotics** (Feb. 2024 – Present): Developed a prototype gaze-driven navigation interface in which a user looks at a point in the robot's camera feed and the robot moves there (or to the nearest reachable position), aimed at people with severe motor impairments. Designing a passive, hands-free error-detection system (a "hands-free e-stop") that uses a brain–computer interface (BCI) to detect error-related potentials when the user perceives a robot error; conducted and analyzed a 10-participant pilot study, and now redesigning the protocol based on the findings.
- **Multi-robot motion planning** (Sept. 2025 – Present): Developed a scalable multi-robot motion planning method that coordinates robots through discrete search over a workspace decomposition, improving planning time by up to an order of magnitude (WAFR 2026); contributed to hypergraph-based multi-robot motion planning (IEEE RA-L 2026) and Lazy-DaSH, a lazy hypergraph-based approach to multi-robot task and motion planning (IEEE T-RO 2026).
- **Insect-scale robot navigation** (Feb. 2021 – 2023): Developed a sampling-based motion planner that incorporates the robots' hopping dynamics to plan feasible paths for a team of low-power, insect-scale robots monitoring crop fields under uncertainty; validated in simulation as part of an interdisciplinary collaboration.

Summer Researcher May 2020 – Aug. 2020
Parasol Lab, University of Illinois Urbana-Champaign · Advisor: Prof. Nancy M. Amato *Champaign, IL*
- Compiled an annotated bibliography on planetary path planning and navigation and presented the findings at the UIUC Summer Pre-Doctoral Institute.

Research Assistant Feb. 2019 – July 2020
Biorobotics Lab, University of Wisconsin–Milwaukee · Advisor: Dr. Mohammad H. Rahman *Milwaukee, WI*
- Developed an adaptive coordinated control system, in MATLAB Simulink and C/C++, for a team of three wheeled omnidirectional mobile manipulators carrying rigid objects along a predefined trajectory.

Technology Lead, RockSat-C Apr. 2018 – June 2020
Students for the Exploration and Development of Space (SEDS), UW–Milwaukee *Milwaukee, WI*
- Developed a sounding-rocket science payload to measure how well different materials shield against harmful radiation in space; the payload launched into space from NASA's Wallops Flight Facility on June 20, 2019.

Publications

Journal Articles

- S. Lee, J. D. Motes, **I. Ngui**, M. Morales, and N. M. Amato, "Lazy-DaSH: A Lazy Approach for Hypergraph-Based Multi-Robot Task and Motion Planning," *IEEE Transactions on Robotics*, 2026.
- C. McBeth, J. D. Motes, **I. Ngui**, M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning Using Workspace Guidance-Informed Hypergraphs," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4929–4936, Apr. 2026. doi: 10.1109/LRA.2026.3666398.

Conference Papers

- **I. Ngui**, C. McBeth, A. Santos, G. He, K. J. Mimnaugh, J. D. Motes, L. Soares, M. Morales, and N. M. Amato, "ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality," in *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)*, Goiânia, Brazil, Sept. 30 – Oct. 2, 2026. arXiv:2510.02464.
- **I. Ngui**, C. McBeth, J. D. Motes, M. Morales, and N. M. Amato, "Scalable Multi-robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement," in *World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, Oulu, Finland, June 15–17, 2026. arXiv:2605.20395.

Poster Presentations

- **I. Ngui**, C. McBeth, G. He, A. Santos, L. Soares, M. Morales, and N. M. Amato, "Extended Reality System for Robotic Learning from Human Demonstration," *IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, Saint-Malo, France, Mar. 2025. arXiv:2409.12862.
- C. McBeth, **I. Ngui**, G. He, A. Santos, L. Soares, M. Morales, and N. M. Amato, "XR-Based System for Demonstration Collection in Human-Robot Interaction," *IEEE International Conference on Robotics and Automation (ICRA) 40th Anniversary Celebration*, Rotterdam, Netherlands, Sept. 2024.
- **I. Ngui**, S. Lee, J. Motes, M. Morales, and N. M. Amato, "A Hierarchical Approach to Workstation-based Task Allocation and Motion Planning," *Robotics & AI in Future Factory Workshop, IROS 2023*, Detroit, MI, Oct. 2023.
- **I. Ngui**, "Trajectory Tracking Control of Three Wheeled Omnidirectional Mobile Robots," *Baylor University McNair Research Conference*, Waco, TX, Sept. 2019. Advisor: Dr. Mohammad H. Rahman.
- K. Krause, **I. Ngui**, W. Roth, and D. Gomez, "Students for the Exploration and Development of Space," *UW System Symposium*, Green Bay, WI, Apr. 2019.

Oral Presentations

- **I. Ngui**, C. McBeth, J. D. Motes, M. Morales, and N. M. Amato, "Scalable Multi-robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement," *World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, Oulu, Finland, June 2026.
- **I. Ngui**, "Planetary Path Planning," *University of Illinois Urbana-Champaign Summer Pre-Doctoral Institute*, Champaign, IL, Aug. 2020. Advisor: Prof. Nancy M. Amato.
- **I. Ngui**, "Trajectory Tracking Control of Three Wheeled Omnidirectional Mobile Robots," *Baylor University McNair Research Conference*, Waco, TX, Sept. 2019; also presented at the *28th Annual McNair Research Presentations*, Milwaukee, WI, Aug. 2019. Advisor: Dr. Mohammad H. Rahman.

Teaching Experience

Teaching Assistant, CS 441: Applied Machine Learning — Fall 2025, Spring 2026
University of Illinois Urbana-Champaign — *Champaign, IL*

Teaching Assistant, CS 491: Professional Development Seminar — Fall 2020
University of Illinois Urbana-Champaign · Instructor: Prof. Nancy M. Amato — *Champaign, IL*

Fellowships & Awards

Golub CS Fellowship, University of Illinois Urbana-Champaign, 2020 — $8,000
Summer Pre-Doctoral Institute Research Grant, University of Illinois Urbana-Champaign, 2020 — $2,500
GEM University Fellowship, 2020
McNair Scholar Summer Research Grant, 2019 — $2,800
Support for Undergraduate Research Fellows (SURF) Award, 2019 — $1,750

Open-Source Software

ERUPT – Extended Reality Universal Planning Toolkit · github.com/parasollab/erupt — 2023 – Present

- XR toolkit for interactive robot motion planning and robot learning with ROS 2: edit scenes by hand, preview robot trajectories in VR/AR, then run them on real robots.

Open Parasol Planning Library (Open PPL) · github.com/parasollab/open-ppl — Feb. 2021 – Dec. 2025

- Helped open-source the Parasol Lab's C++ motion planning library for robotics and other planning applications.

Community & Leadership

Volunteer Coding and Robotics Instructor, Central City Cyberschool, Milwaukee, WI — 2019
Mentor, UWM Black Male Summit, Milwaukee, WI — 2019

Technical Skills

Languages: C/C++, Python, Java, C#, Julia
Tools: ROS, ROS 2, PyTorch, TensorFlow, OMPL, MATLAB, Simulink, Unity